# 1 Code Execution Procedure

This project contains four C++ programs for rainfall classification using logistic regression. All programs use the same CSV dataset generated by the Python file `datapy.py`.

## 1.1 Project Files

Table 1: Project files and their purpose

| File Name | Description |
|---|---|
| `datapy.py` | Python file used for CSV dataset generation |
| `seq1.cpp` | Sequential logistic regression code |
| `popen.cpp` | OpenMP parallel logistic regression code |
| `hybrid.cpp` | Hybrid OpenACC + MPI implementation |
| `final.cpp` | Final OpenACC + MPI implementation |

## 1.2 Step 1: Generate the CSV Dataset

```
python datapy.py
```

This command generates the CSV dataset used by all four C++ codes.

## 1.3 Step 2: Run Sequential Code

```
g++ seq1.cpp -o seq1
seq1.exe
```

## 1.4 Step 3: Run OpenMP Code

```
g++ -fopenmp popen.cpp -o popen
popen.exe
```

To set number of OpenMP threads:

```
set OMP_NUM_THREADS=4
popen.exe
```

## 1.5 Step 4: Run Hybrid OpenACC + MPI Code

```
mpic++ -acc hybrid.cpp -o hybrid
mpirun -np 4 hybrid.exe
```

## 1.6 Step 5: Run Final OpenACC + MPI Code

```
mpic++ -acc final.cpp -o final
mpirun -np 4 final.exe
```

## 1.7 Execution Summary

Table 2: Compilation and execution commands

| Implementation | Compile Command | Run Command |
|---|---|---|
| Sequential | `g++ seq1.cpp -o seq1` | `seq1.exe` |
| OpenMP | `g++ -fopenmp popen.cpp -o popen` | `popen.exe` |
| Hybrid OpenACC + MPI | `mpic++ -acc hybrid.cpp -o hybrid` | `mpirun -np 4 hybrid.exe` |
| Final OpenACC + MPI | `mpic++ -acc final.cpp -o final` | `mpirun -np 4 final.exe` |